

Enhanced Brain Stroke Prediction: An Ensemble of Random Forest, Logistic Regression and XGBoost

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Abstract—Stroke, a leading cause of disability and mortality globally, necessitates early prediction and intervention to mitigate its devastating effects. Recognizing the challenges faced by stroke survivors, the focus shifts to the crucial role of rehabilitation and lifestyle changes in optimizing recovery and quality of life. Machine learning (ML) algorithms emerges as a powerful tool for predicting stroke risk and enhancing patient outcomes. ML algorithms, by analysing extensive datasets, can identify patterns and early warning signs, enabling more personalized treatment plans. To enhance prediction accuracy, an ensemble model is proposed, leveraging Random Forest (RF), Logistic Regression (LR) and XGBoost techniques. The ensemble model exhibits an impressive accuracy rate of 97.4%, exceeding other models.

Keywords—brain stroke, ensemble model, stroke prediction, healthcare AI, predictive analytics

I. INTRODUCTION

Stroke disease occurs when there is blockage in the flow of blood causing the brain tissues from getting necessary oxygen flow. It can lead to sever impairment and also lead to death [1]. The impact of a stroke can be devastating, manifesting symptoms ranging from temporary paralysis and speech difficulties to permanent loss of movement or cognitive impairment. Early prediction and intervention are crucial for improving stroke outcomes and reducing the burden of this deadly disease [2]. Timely interventions are critical in stroke management, enabling medical professionals to administer treatments such as clot-busting drugs and mechanical thrombectomy, which prove most effective when applied within the first few hours of symptom onset. The ability to identify individuals at high risk for stroke and implement preventative measures, such as lifestyle changes and medication, can ultimately save lives and mitigate the devastating effects of this condition [3]. The effects of stroke can be profound, disrupting an individual's normal daily activities and independence. Depending on the severity and location of the brain damage, stroke survivors may experience a range of symptoms, from physical impairments such as paralysis, weakness, or numbness, to cognitive deficits including memory loss,

difficulty with speech, or impaired reasoning [4]. These challenges can make it difficult for individuals to perform basic tasks, engage in social interactions, or return to work, leading to a loss of independence and quality of life. While stroke cannot be completely cured, early intervention and rehabilitation are crucial for maximizing recovery and quality of life for stroke survivors. The goal of stroke treatment is to restore blood flow to the brain as quickly as possible to minimize damage and prevent further complications. Depending on the type of stroke, treatment options may include clot-busting drugs, mechanical thrombectomy, or surgery to repair damaged blood vessels [5]. Rehabilitation programs focus on improving physical strength, mobility, speech, and cognitive function to help individuals regain independence and achieve the highest level of functioning possible. With a multidisciplinary approach that emphasizes prevention, acute care, rehabilitation, and ongoing support, individuals affected by stroke can achieve meaningful improvements in their health and well-being [6]. Machine learning has emerged as a powerful tool for predicting stroke risk and improving patient outcomes. By analysing vast amounts of data, ML algorithms can uncover patterns and trends that may elude healthcare professionals [7]. These algorithms assess factors such as blood pressure, cholesterol levels, medical history, and lifestyle habits to accurately predict an individual's risk of experiencing a stroke. ML models can also be trained to recognize early warning signs of a stroke based on symptoms reported by patients, such as sudden weakness or numbness in the face, arm, or leg. By leveraging ML technology, healthcare providers can develop more personalized and targeted treatment plans for high-risk individuals, ultimately leading to better stroke prevention strategies and outcomes [8]. ML holds the potential to revolutionize stroke prediction and management, offering a valuable tool in the fight against this deadly disease. To enhance stroke prediction accuracy, an ensemble algorithm is proposed, combining the strengths of three popular machine learning techniques: RF, LR and XGBoost (Extreme Gradient Boosting). The ensemble learning

approach facilitates the creation of a more robust model by combining predictions from these three models. The ensemble model demonstrates an impressive accuracy rate of 97.4%, surpassing the performance of individual models [9]. The Random Forest algorithm employs decision trees and random sampling techniques to identify critical features and mitigate overfitting. SVM, a potent technique, handles nonlinear data by determining the optimal hyperplane that maximizes the margin between classes. XGBoost, a gradient boosting algorithm, combines weak learners iteratively, improving predictions with each iteration. The integration of these techniques results in a more robust and accurate model for stroke prediction, contributing to enhanced prevention and management strategies for patients at high risk.

II. LITERATURE SURVEY

In this section, we conduct an analysis and review of previously published papers focusing on the prediction of stroke types through the application of diverse machine learning approaches. The paper [10] involves stroke prediction through an enhanced random forest algorithm, which helps analyse stroke risk levels. Compared to previous methods, this approach seems to be more effective. However, it should be noted that this study is restricted to a limited stroke category and may not be applicable to new stroke types in Realtime. This paper [11] examines ML approaches for Stroke Prediction by analysing previously published works. Predominantly focusing on mortality rates and functional outcomes, the commonly utilized techniques include RF, SVM, DT and neural networks. Despite these efforts, a lack of standardized reporting in medical sector tools has hindered practical utility, with few predictors and classifiers proving effective. In the paper [12], various classifiers are applied and compared to diagnose the Stroke dataset effectively. The method presents the effectiveness of the Ensemble approach in enhancing prediction accuracy, as evidenced by experimental results. This technique relies on supervised learning, combining predictions from multiple algorithms for estimation. The research in [13] employs LR as a key method for binary classification, appreciating its simplicity, ease of implementation, and interpretability. By conducting feature importance analysis, the study identified age, hypertension, and average glucose level as the most influential factors. This study's application of logistic regression and feature analysis achieved an impressive 90% accuracy rate, showcasing its potential in risk prediction and classification tasks. This study [14] introduces a predictive model designed to anticipate the occurrence of brain strokes in individuals. Evaluation of the model's performance utilizes various metrics. A thorough analysis compares the outcomes generated by multiple machine learning algorithms incorporated into the study. Results indicate that various algorithmic approaches demonstrate efficacy in predicting brain strokes among patients. The logistic regression algorithm emerges as the top-performing method, achieving an accuracy rate of 95.02% when compared against alternative machine learning techniques. This research [15] investigated the effectiveness and characteristics of six machine learning models, all of which utilize boosting techniques and ensemble learning methods for predicting brain strokes. The models examined multiple boosting models and two datasets with similar attributes

were employed in the analysis. The findings suggested that a stacked model demonstrated strong performance in identifying optimal mapping functions for stroke prediction, achieving an impressive accuracy of 97.88%. This study provides valuable insights into the development of a generalized prediction model for early detection of brain strokes using boosting-based stacking techniques. This research paper [16] focuses on early stroke prediction using various machine learning approaches. Ten classifiers including LR, DT Classifier, and XGBoost Classifier, were trained on the attributes. The study achieved an impressive 97% accuracy, with the weighted voting classifier outperforming individual methods. The weighted voting model exhibited the lowest false positive and false negative rates, making it an ideal tool for physicians and patients in the early detection and prescription of potential strokes. This paper [17] introduces a hybrid machine learning approach for predicting cerebral stroke in clinical diagnosis, addressing challenges such as incomplete physiological data and class imbalance. The method involves two key steps: first, utilizing RF regression to identify missing values and second employing an automated tuning of parameters using a deep learning model for stroke prediction on an imbalanced dataset. The specified approach demonstrates a notable improvement with a false negative rate of 19.1%, reduced by an average of 51.5% compared to traditional methods.

III. METHODOLOGY

In developing an effective stroke prediction model, the choice of machine learning algorithms is pivotal. The ensemble approach, combining the strengths of Random Forest, Logistic Regression, and XGBoost, was meticulously chosen to harness the unique advantages each algorithm offers. Predicting the stroke at early stage can to save a human life with early treatment.

A. Random Forest

RF employs a group of decision trees, each constructed employing a randomly selected features of the input parameters. This technique allows the model to capture complex relationships within the dataset while mitigating the risk of overfitting [18]. The adaptability of RF to both continuous and categorical variables make it a robust choice for our ensemble. The fundamental equation for a decision tree is Eqn 1.

$$Y = f(X; \theta) \quad (1)$$

B. Logistic Regression

LR is a ML based classification model that is used to predict the probability of a categorical dependent variable. LR a statistical technique used for binary classification tasks, where the objective is to determine the probability that a given data belongs to a particular category. It models the relationship among one or more predictor variables and the probability of the binary outcome using the logistic function. During training, the model learns optimal coefficients that minimize the difference between predicted probabilities and actual outcomes. LR is valued for its

simplicity, interpretability, and effectiveness in various domains, including healthcare, marketing, and finance.

C. XGBoost (Extreme Gradient Boosting)

XGBoost is a variant of gradient boosting model, iteratively combines weak learners, continually refining predictions with each iteration. This iterative improvement mechanism allows the model to adapt and enhance its accuracy over multiple rounds. Eqn 3 is the fundamental equation driving XGBoost's learning process.

$$obj(\theta) = \sum_{i=1}^n L(y_i, f_i) + \sum_{k=1}^K \Omega(f_k) \quad (3)$$

The ensemble model aggregates the predictions of these individual algorithms, namely RF, LR and XGBoost leveraging their diverse approaches to enhance overall accuracy. The majority voting method is used to create the ensemble model. This combination results in a more comprehensive and reliable stroke prediction system, capable of identifying subtle patterns and nuances within complex medical datasets that may go unnoticed when relying on a single algorithm [19]. By combining the strengths of multiple models, the ensemble approach provides a more robust framework for stroke prediction, particularly in scenarios where the data is noisy or when different algorithms excel in different aspects of prediction.

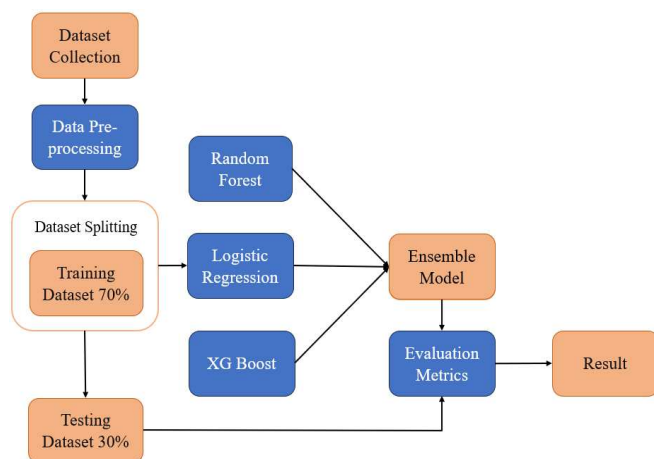


Figure 1 Overview of proposed method

IV. EXPERIMENTAL SETUP

To efficiently predict brain strokes and achieve the highest accuracy, a benchmarked dataset is essential. The dataset utilized for developing the ensemble algorithm was obtained from Kaggle, comprising 5110 rows and 11 columns [20]. The attributes describe demographic and health-related information of individuals in a dataset. They include unique identifiers, demographic factors such as gender, age, marital status, and residence type, as well as health indicators and history of stroke. These attributes provide a comprehensive overview of the individuals' backgrounds and health conditions, which can be analysed for various purposes such as healthcare management and predictive modelling. Figure 2 illustrates the percentage of individuals with strokes in the dataset, while Figure 3 depicts the relationship between BMI and stroke incidence. Also, Figure 4 represents the occurrence of stroke among individuals with heart disease and hypertension.

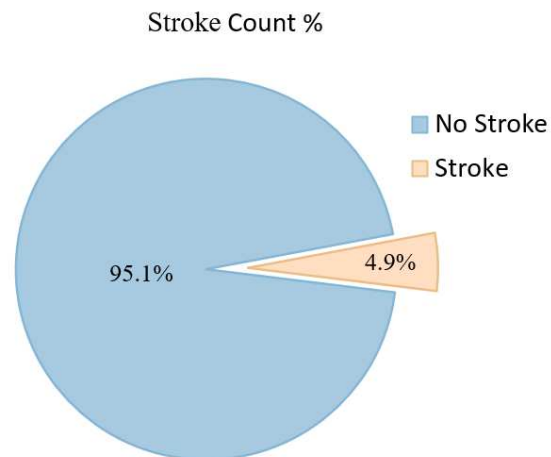


Figure 2 Percentage of Stroke count on Dataset

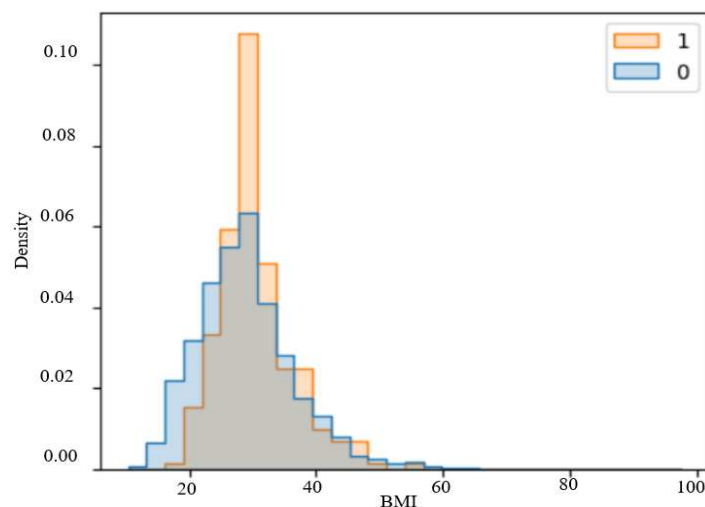


Figure 3 Stroke on BMI

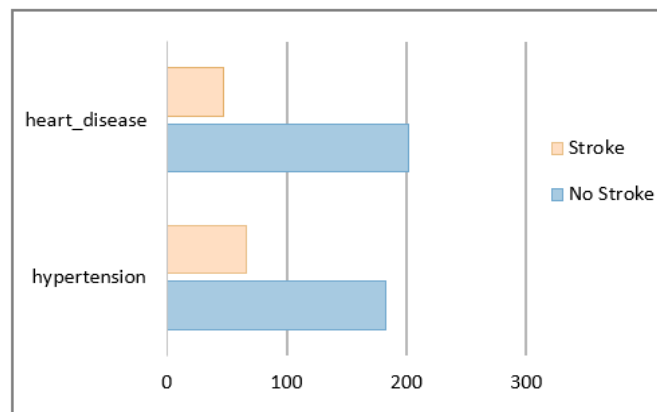


Figure 4 Stroke among individuals with heart disease and hypertension

Before proceeding with the dataset, unwanted noise and outliers should be removed to ensure the efficiency of the algorithm [21]. Once the dataset is cleaned, it is then split into training and testing data.

Each machine learning algorithm—Random Forest, Logistic Regression, and XGBoost—underwent a rigorous evaluation to discern its individual predictive capabilities for stroke prediction [22]. The assessment was conducted using a comprehensive set of metrics on different classifiers and the proposed ensemble method.

RF with its ensemble of decision trees demonstrated a commendable performance in predicting strokes[23]. The algorithm's adaptability to both continuous and categorical variables proved crucial in capturing intricate relationships within the dataset. The evaluation yielded an accuracy rate of 94.7%, emphasizing its proficiency in identifying stroke risk.

Logistic Regression is renowned for its simplicity and interpretability has emerged as a stalwart in discerning the intricate relationships within stroke prediction data. By estimating the probability of stroke occurrence based on a set of input features, Logistic Regression adeptly navigated the data landscape, revealing subtle yet crucial patterns. Its straightforward approach in modelling the log odds of the response variable allowed for a clear understanding of the predictive factors contributing to stroke risk. Evaluation metrics, encompassing accuracy, precision, recall, and F1 score, collectively affirmed Logistic Regression's efficacy in capturing the nuanced dynamics underlying stroke susceptibility.

XGBoost, leveraging its gradient boosting technique, iteratively refined predictions with each iteration, showcasing a continual improvement in accuracy. The algorithm's ability to integrate weak learners effectively contributed to its predictive prowess in the context of stroke prediction. Evaluation metrics revealed 92.8%, reaffirming XGBoost's role as a valuable component in the ensemble model.

Table I. Metrics analysis of algorithms

Algorithm	Accuracy	Recall	Precision	F1 Score
Random Forest	94.7%	93%	94.2%	95.6%
Logistic Regression	89.6%	92%	90.4%	93.2%
XGBoost	92.8%	90.2%	92.8%	94%
Ensemble Algorithm	97.4%	96.7%	98.1%	95.8%

After individually assessing each algorithm's performance, a comparative analysis was undertaken as shown in Table I. The graphical representation of the comparison is illustrated in Figure 5.

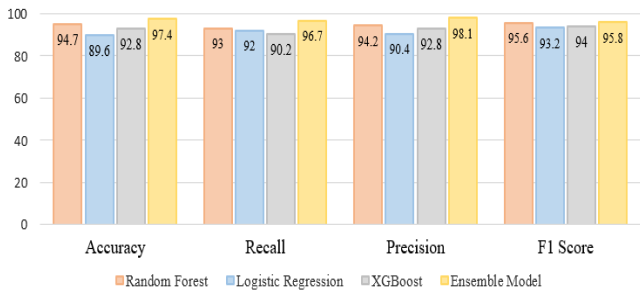


Figure 5 Comparison of Ensemble Model with individual models

It is clear that the ensemble method outperformed the approaches and is capable of predicting stroke effectively. The ensemble model, a synthesis of the predictions from these distinct algorithms, exhibited an improvement in predictive accuracy. Surpassing the individual models, the ensemble achieved an overall accuracy rate of 97.4%, highlighting the synergistic benefits derived from the combination of diverse algorithms.

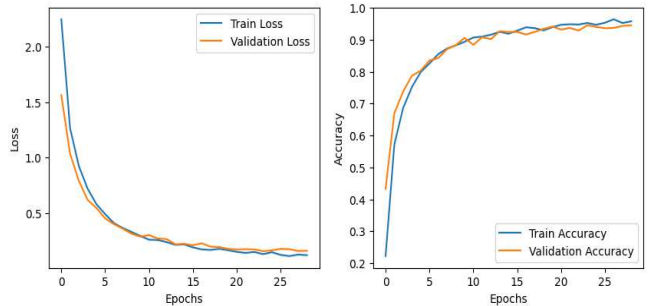


Figure 6 Training vs Validation Performance

Figure 6. presents the performance of proposed approach in terms loss and accuracy for considered training and test set. It is clearly evident that the ensemble model is capable of predicting the stroke disease accurately for test set.

V. CONCLUSION

This study focused on combining different algorithms (RF, LR and XGBoost) to make a better tool for predicting strokes. The performance of each algorithm was measured. But the most significant improvement occurred when all three were combined into an ensemble model. This combined approach greatly improved our ability to predict strokes accurately. The overall accuracy rate of 97.4% for the ensemble model was better than any single algorithm. This shows that using a mix of these algorithms works better and could be a more powerful tool for predicting strokes precisely. This study highlights the effectiveness of combining approaches in healthcare analytics, providing a stronger method for accurate stroke predictions and potentially leading to better early detection and intervention strategies

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